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{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:45:34",
  "variant" : 1,
  "subjectName" : "computer-networks",
  "categories" : [ "Routing Protocols", "Network Security Basics", "IP Addressing & Subnetting" ],
  "questions" : [ {
    "question" : "What metric parameter does the Routing Information Protocol (RIP) utilize to determine the best path?",
    "answers" : [ "Total circuit delay latency", "Link cost speed calculations", "Bandwidth capacity", "Hop count" ],
    "correctIndex" : 3
  }, {
    "question" : "Which of the following routing protocols operates as an Exterior Gateway Protocol (EGP) to handle internet traffic?",
    "answers" : [ "Border Gateway Protocol (BGP)", "Routing Information Protocol (RIP)", "Open Shortest Path First (OSPF)", "Enhanced Interior Gateway (EIGRP)" ],
    "correctIndex" : 0
  }, {
    "question" : "What algorithm forms the mathematical engine used by OSPF to compute shortest path trees?",
    "answers" : [ "Dijkstra's Algorithm", "Floyd-Warshall Routine", "Banker's Algorithm", "Bellman-Ford Algorithm" ],
    "correctIndex" : 0
  }, {
    "question" : "What is the role of a default gateway entry inside a client machine's routing profile?",
    "answers" : [ "The destination path used for traffic traveling outside the local network subnet boundary", "The local database server recording updates", "The backup DNS server utilized when primary queries fail", "The physical hardware switch port connected to the line" ],
    "correctIndex" : 0
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    "question" : "How does Asymmetric Encryption handle session key setup operations?",
    "answers" : [ "It restricts traffic lanes to specific local MAC configurations", "It uses a public key to encrypt data and a matching separate private key to decrypt it", "It hashes password strings using random salt variables", "It relies on a single shared symmetric key rotated periodically" ],
    "correctIndex" : 1
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    "question" : "Which protocol encapsulates web traffic inside an encrypted SSL/TLS session framework layer?",
    "answers" : [ "SFTP", "HTTP", "HTTPS", "SNMP" ],
    "correctIndex" : 2
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"question" : "What type of intrusion occurs when an attacker intercepts and modifies
communications between two endpoints?",
"answers" : [ "SQL Injection (SQLi)", "Denial of Service (DDoS)", "Man-in-the-Middle (MitM)
attack", "Cross-Site Request Forgery (CSRF)" ],
"correctIndex" : 2
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"question" : "What type of firewall monitors the state of active connections to determine whether to
block a packet?",
"answers" : [ "Packet Filtering Firewall", "Stateful Inspection Firewall", "Stateless Lookups
Interface", "Application Proxy Gateway" ],
"correctIndex" : 1
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"question" : "If a host configuration has an IP of 192.168.1.50 with a subnet mask of /26, how
many hosts are available?",
"answers" : [ "62 hosts", "64 hosts", "30 hosts", "126 hosts" ],
"correctIndex" : 0
}, {
"question" : "Which mechanism enables multiple private local network clients to share a single
public IP mapping parameter?",
"answers" : [ "Dynamic Host Configuration (DHCP)", "Domain Name Resolution (DNS)", "Network
Address Translation (NAT)", "Virtual Local Architecture (VLAN)" ],
"correctIndex" : 2
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